RouteWise: Shipment Route Recommendation under Cost, Time and Risk

Aiqon AI Internship take-home · technical report · code, data and tests in the accompanying repository

1. Problem and definition of optimal

A planner has one shipment and 3–5 candidate multi-leg routes. The cheapest route is usually slow or unreliable. The fastest is usually expensive or runs through a high-risk chokepoint. No route is optimal in the abstract, so RouteWise starts from an explicit definition:

The optimal route is the one that satisfies the shipment's hard constraints (deadline, budget, cargo risk limit) and, among the non-dominated routes that remain, has the lowest weighted trade-off score across cost, expected transit time, probability of missing the deadline, and route risk. The weights are set by shipment priority and cargo type.

Each part of the definition is a separate design choice. **Constraints are not traded away:** a cheap route that breaches a hard deadline is *invalid*, not merely low-scoring. No weighting should be able to buy it back. **Pareto filtering** guarantees the system never recommends a route that another route beats on every criterion. **Weighted scoring** settles the remaining trade-offs. Of the methods considered, it is the easiest to explain to an operator, and the brief judges explainability explicitly. **Context-dependent weights** reflect the fact that urgent pharma and bulk steel should not be optimised the same way.

2. Decision framework

`shipment → validate → hard constraints → Pareto filter → Monte Carlo → score → rank → explain → confidence`

- **Four criteria, all lower-is-better.** C = total cost. T = expected transit = base + P(delay) × mean delay. R = P(arrive after deadline), from simulation. K = 0.5 × congestion + 0.5 × geopolitical risk.
- **Hard constraints.** P90 transit ≤ deadline (when the deadline is hard); cost ≤ budget cap; geopolitical risk ≤ cargo limit. The deadline is checked against **P90, not the mean**: a route that arrives on time on average but misses 40% of the time is not compliant. If nothing passes, the system returns the *least-violating* route (smallest normalised total violation), flags it NON-COMPLIANT and lists every failed constraint. It always returns an answer.
- **Uncertainty (Monte Carlo).** 1,000 seeded draws per route. Each draw makes a Bernoulli delay decision; if delayed, the duration is lognormal with its mean pinned to the route estimate. Outputs are P50, P90, P(miss) and expected days late. This makes tail risk a decision input rather than a footnote. A route with a good mean but a fat tail is penalised on R and can fail the P90 check.
- **Scoring.** Each criterion is min-max normalised *within the shipment's own option set*, so scores are relative to the real alternatives. A criterion whose spread is under 2% of its minimum is treated as flat, so rounding-level differences cannot decide. Score = Σ weight × normalised value.
- **Explanation.** Deterministic templates, **not an LLM**, so numbers cannot be hallucinated and are testable. A test asserts that every dollar figure printed traces back to the data. Each explanation gives: a headline; the 2–3 criteria with the largest weighted advantage over the runner-up, with real deltas; the trade-off accepted; one sentence each versus the cheapest and fastest routes; eliminated routes with their reason ("dominated by R-x" or the violated constraint); confidence; and warnings.

Profile	Cost	Time	Reliab.	Risk	Applied when
urgent	0.15	0.40	0.30	0.15	priority = urgent
standard	0.35	0.25	0.25	0.15	priority = standard
budget	0.60	0.10	0.15	0.15	priority = budget
critical	0.20	0.15	0.30	0.35	perishable / fragile / high-value (overrides)

Table 1. Weight profiles (config/weights.yaml). These are transparent, editable defaults, not learned values. Section 4 tests whether the conclusions depend on them.

Approaches not used. *A learned ML model:* there are no real outcome labels, so a model would only learn the synthetic generator, and it would be unexplainable. *ILP or OR-Tools:* routes are pre-enumerated (3–5 per shipment), so exhaustive scoring is already exact and an optimiser adds nothing. *Pure lexicographic rules:* easy to explain, but they ignore the size of a trade-off. A 1% cost saving should not outweigh a 30-point drop in reliability.

3. Synthetic data and how the trade-offs were built

The data has 14 ports (Mumbai, Colombo, Jebel Ali, Singapore, Port Said, Jeddah, Cape Town, Rotterdam, Hamburg, Shanghai, Busan, Los Angeles, New York, Santos), 34 sea/air/rail legs, and 10 lanes with 4–6 path templates each: direct, one or two transshipments, a Cape corridor instead of Suez, and sea-air hybrids. From these the generator builds **40 shipments and 163 routes** (3–5 per shipment) with seed 42, reproducible byte-for-byte (tested). Route attributes are aggregated from legs: cost = leg freight × volume (discounted 0.72× per transshipment, reflecting feeder consolidation) + handling; transit = sailing days + dwell that scales with hub congestion; delay probability = $1 - \prod(1 - p_{leg})$; geopolitical risk = **max** over hubs (one chokepoint can dominate a route); congestion = mean over hubs. All estimates carry ±5% noise.

The trade-off structure is built in deliberately and then measured. Cheaper routes transship more, so they are slower and more congested. Direct Suez routes are fast but carry 0.78 geopolitical risk. The Cape corridor is safe but slow. Air hybrids are fast, reliable and expensive. Within each shipment's option set, the cost-vs-transit correlation is **-0.37**. This is the correlation that matters, because the decision only compares routes within a shipment. The pooled cross-lane figure is confounded by lane length.

18 routes (11%) are Pareto-dominated, which exercises the filter. The data also includes injected cases: 4 *trap* shipments where the cheapest route misses a hard deadline, 4 *near-ties*, and 2 shipments with *no feasible route*. Deadlines come from the **simulated** P90 of the shipment's fastest relevant route (×1.1 urgent, ×1.3 standard, ×1.6 budget). The generator calls the same simulator the engine uses. In an earlier version that approximated P90 instead, deadlines and constraints disagreed, and 8 shipments became accidentally infeasible.

4. Evaluation

There are no ground-truth labels, so RouteWise is evaluated by comparing strategies on shared metrics and checking expected behaviours. To keep the evaluation from being circular, the money metric, **expected landed cost** (freight + inventory holding over simulated transit + late penalty × expected days late), is *not* what the scorer optimises. **Worst-case regret** is the largest normalised regret across the four criteria, measured against the routes the operator is *allowed* to book. Regret against a route that breaches a hard constraint is not regret anyone can act on. Values above 1 mean the pick is worse than every compliant option on some criterion.

Strategy	Mean cost	E[transit]	P(miss)	Worst-case regret	Landed cost	Constraint violations	Pareto-optimal
A1 RouteWise	**$15,596**	**26.5 d**	**6.0%**	**0.62**	**$23,004**	**5%**	**100%**
A2 Ablation: no constr./Pareto	$13,708	28.6 d	11.2%	0.68	$20,840	15%	100%
B4 Equal weights	$16,795	23.4 d	8.5%	3.62	$22,674	32%	100%
B2 Always fastest	$21,902	18.5 d	5.7%	3.69	$26,120	32%	100%
B3 Always lowest risk	$17,731	27.3 d	26.4%	8.67	$26,924	28%	100%
B5 Random (100 draws)	$15,117	30.0 d	23.6%	11.00	$23,349	40%	87%
B1 Always cheapest	$10,188	32.6 d	22.7%	18.19	$18,472	55%	100%

Table 2 (E1). All strategies on 40 shipments. A1's 5% violation rate is exactly the 2 deliberately infeasible shipments, each returned with the NON-COMPLIANT flag.

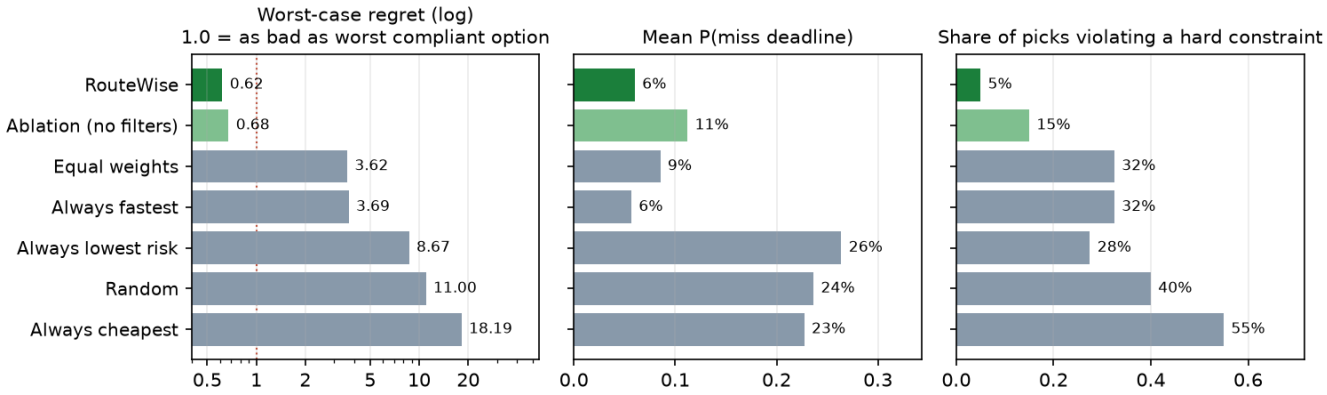


Figure 1. RouteWise vs baselines on 40 shipments; lower is better in every panel. Dotted line at regret 1.0 = as bad as the worst compliant option on some criterion.

Findings, including the ones that do not favour RouteWise.

- **Best balance.** RouteWise has the lowest worst-case regret (0.62). Always-fastest scores 3.69 and always-cheapest 18.19. As expected, it is rarely the best on any single metric, but it never makes a lopsided pick. 100% of its recommendations are Pareto-optimal.
- **Cost premium, stated openly.** Against always-cheapest, the deadline-miss rate falls from 22.7% to 6.0%, at a **53% higher mean freight cost**. Always-fastest has a slightly lower miss rate (5.7%) but costs $6,306 more per shipment and violates a hard constraint in 32% of cases, usually the cargo geopolitical-risk limit.
- **RouteWise does not win on expected landed cost.** B1 ($18,472) and A2 ($20,840) beat it ($23,004). Most of the gap comes from the hard constraints, and that is intended: the yardstick prices lateness linearly and gives no value to a geopolitical-risk limit. The constraints exclude cheap options a money-only view would accept. Comparing A1 with the ablation A2 isolates the cost of the constraints: $1,888 per shipment buys a fall in violations from 15% to 5% and in miss rate from 11.2% to 6.0%.
- **E2, profiles behave as intended.** With all 40 shipments forced through each profile, mean transit is 23.5 / 27.0 / 30.9 d and mean cost is $19,123 / $15,734 / $13,671 for urgent / standard / budget. Both are monotone, as required.
- **E3, noisy estimates.** When the system sees attributes with 15% noise and is scored against the clean truth, its decision changes in 30% of shipments. The median regret against the clean-data oracle is $1,511 (mean $4,741, skewed by a few large misses). This is the most important limitation: input quality, not the weights, is the dominant source of decision error.

E4, sensitivity and confidence. Each shipment's weights were perturbed by U(0.8, 1.2) and renormalised, 500 times. The same route wins in **96.7%** of runs on average, a dataset-level flip rate of 3.3%. The cost weight causes the most flips (mean 5.5% when perturbed alone, versus 2.6% for time). That fits the data, where cost is the widest-spread criterion. Confidence is **High** when stability ≥ 90% and margin ≥ 0.05, **Medium** at 70–90%, and **Low** below 70% (shown as "close call, review manually"). Result: 31 High, 8 Medium, 1 Low. All 4 injected near-ties are below High (tested). Developing this exposed a real flaw. Min-max normalisation stretches a 1% raw gap to a 0→1 score gap, so near-identical routes showed a large margin *and* 100% stability, and were labelled

High. A raw-value closeness check now caps such cases at Medium. A one-at-a-time weight sweep reports **switching points**. For S013, for example, the recommendation changes at cost weight ≈ 0.46. This tells an operator directly how strongly a priority would have to change to change the answer.

E5, a trap case (S004, budget priority, cost weight 0.60). The engine output, abridged:

```
Recommended: R-S004-1 via SHA > RTM. Cost $2,924 | E[transit] 28.8d (P90 31.3d)
  1% chance of missing the 36.0d hard deadline.
vs cheapest (R-S004-2): we pay $1,310 more (+81.1%); in exchange deadline-miss
  risk falls from 100% to 1%, it arrives 17.4d sooner, route risk 0.67 -> 0.45.
  The cheapest option is not allowed: P90 transit 49.5d exceeds hard deadline
  36.0d by 13.5d.
Confidence: High (same route wins in 100% of 500 perturbed runs).
```

The full near-tie and no-feasible-route case studies are in `reports/e5_case_studies.md`.

5. Verification

There are 92 tests, with 92% line coverage, run in GitHub Actions CI together with ruff. The tests include **property-based tests (hypothesis, 200 random cases each)** for the invariants that make the framework trustworthy: raising a route's cost never improves its rank; adding a dominated route never changes the winner; scaling all costs never changes the winner; and permuting route order never changes the winner. Golden cases cover the behaviours an operator would expect: urgent picks fast, budget picks cheap, fragile cargo avoids the chokepoint, the trap case is avoided, and the fallback works. The PRD success criteria are written as tests, so they are checked rather than claimed. One recommendation, including 1,000 Monte Carlo draws, takes well under 1 s. Output is deterministic, and one command (`make all`) regenerates the data, results and figures.

6. Assumptions

- All costs are in USD per shipment; transit time is in days, door to door. Route attributes are estimates available at decision time, not guarantees.
- Leg delays are independent. When a delay happens, its duration is lognormal (right-skewed).
- Geopolitical risk is worst-link (max over hubs); congestion is the mean over hubs.
- Priority and cargo type are known inputs. Capacity is not modelled: every listed route can take the shipment.
- Hard deadlines are checked at P90. Cargo geopolitical-risk limits are safety policy and are never relaxed. Where every route on a lane breaches the limit, the shipment is booked under a recorded waiver, never silently passed as compliant.
- The synthetic data is plausible, not calibrated to real freight rates. Conclusions concern the decision procedure, not absolute costs.

7. Limitations and future work

Limitations. The synthetic data may miss real correlations; in particular, delays on shared chokepoints are correlated in practice, so the independence assumption understates tail risk on multi-chokepoint routes. The weights are judgement-based, not learned or elicited from operators; sensitivity analysis bounds this but does not remove it. Min-max normalisation depends on which alternatives are on the menu: adding a far-off option can rescale the others. Evaluation proxies come from the model, not real shipment outcomes. E3 shows that 15% input noise changes 30% of decisions, so in practice the quality of the estimates matters more than tuning the scorer.

Future work. Calibrate against real rate and delay histories, including correlated chokepoint shocks. Elicit weights with pairwise comparisons, or learn them from operators' past choices. Feed live congestion and risk indices into the model. Add capacity and schedule constraints. Add a constrained LLM layer that only rephrases the template explanation; it would never be allowed to add or change numbers.